

# Analysis PhD Exam, September 2005

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*Give complete proofs and computations. Be particularly careful to indicate why the most crucial steps in your proof are true. Partial credit will be given where justified.*

1. Let  $\Omega \subset \mathbb{R}^n$  be open. Prove that  $A \subset \mathcal{D}(\Omega)$  is bounded if and only if

$$\sup\{|\Lambda(\varphi)| \mid \varphi \in A\} < \infty,$$

for every  $\Lambda \in \mathcal{D}'(\Omega)$ .

2. Let  $\mu$  be Lebesgue measure on  $\mathbb{R}$  and  $f$  be  $\mu$ -integrable on  $[0, 1]$ . Suppose  $\int_0^r f d\mu = 0$ , for every rational  $r \in [0, 1]$ . Give a complete characterization of  $f$ .
3. Let  $\mathcal{H}$  be a Hilbert space and suppose  $A$  is a bounded operator on  $\mathcal{H}$  such that the range of  $A$  is finite dimensional. Let  $\{y_1, \dots, y_n\}$  be an orthonormal basis for the range of  $A$ . Prove that there exist  $x_1, \dots, x_n \in \mathcal{H}$  such that

$$Ax = \sum_{j=1}^n [x, x_j] y_j,$$

for all  $x \in \mathcal{H}$ .

4. Let  $A$  be a bounded operator on a Hilbert space  $\mathcal{H}$  such that  $A(B_1) = \{Ax \mid x \in B_1\}$  is compact, where  $B_1$  is the closed unit ball of  $\mathcal{H}$ . Prove that, as a map from  $B_1$  with the weak vector topology to  $\mathcal{H}$  with the norm topology,  $A$  is continuous. (Hint: show that the weak and norm topologies on  $A(B_1)$  must coincide under the given hypothesis.)
5. Let  $\mu$  be a positive Radon measure on  $\mathbb{R}^N$  and  $f$  be  $\mu$ -integrable. Show that for each  $\epsilon > 0$  there exist an upper semi-continuous function  $u_\epsilon$  and a lower semi-continuous function  $v_\epsilon$  such that  $u_\epsilon \leq f \leq v_\epsilon$  and  $\int (v_\epsilon - u_\epsilon) d\mu < \epsilon$ .
6. Let  $\mu$  be a finite positive Radon measure on  $\mathbb{R}^N$  and  $\{f_n\}_{n \in \mathbb{N}} \subset L^2(\mu)$  with  $\|f_n\|_2 \leq 1$ . Suppose  $\{f_n\}_{n \in \mathbb{N}}$  converges  $\mu$ -a.e. to  $f$ . Show that  $\int |f_n - f| d\mu \rightarrow 0$ , as  $n \rightarrow \infty$ .
7. Let  $E$  be a topological vector space. Prove that if  $A$  and  $B$  are compact subsets of  $E$ , then so is  $A + B$ .
8. Consider  $X \in \mathcal{D}(\mathbb{R}^2)^*$  defined by

$$\mathcal{D}(\mathbb{R}^2) \ni \phi \mapsto X(\phi) = \int_{\mathbb{R}} \phi(x, -x) dx.$$

Note. So  $X$  is a solution of the PDE  $(\partial_1 - \partial_2)X = 0$  which is not a classical solution.

- (a) Show that  $X$  is a distribution of order zero.
- (b) What is the support of  $X$ ? (Prove it.) Use this to prove that  $X$  is not a regular distribution with continuous coefficient function.
- (c) Show that  $(\partial_1 - \partial_2)X = 0$ .